# OGA · Oriented Geometric Alignment & Adaptive Loss

Gaussian BEV · one independent criterion per task group · optional IoU quality supervision

## Dual-stream supervision: parameter losses + joint geometry

Prediction + ground truth · heatmap, metric offset, log-size, doubled-yaw vector [+ IoU logit]

Heatmap classification
Modified focal loss
all grid cells

Offset regression
Masked Smooth-L1
positive cells

Log-size regression
Masked Smooth-L1
positive cells

Doubled-yaw regression
Masked Smooth-L1
positive cells

Oriented geometry
Lgeo = Lproj + βcorner D
derived from regression

T-SBUW · EMA calibration + learned temperature-softmax task weights
Tasks: cls, offset, size, yaw, geo [+ iou] · M=5 or 6

Repeat independently per group: separate learned task logits and EMA buffers

## Optional IQA · enabled here

Quality target q = stop-gradient MGIoU
BCEWithLogits(iou_logit, q)
reg_mask only · iou_loss_weight=1

## Geometry supervision · positive cells only

Decode predicted and GT corners
size = exp(clamp(log-size, −10, 10))
angle = 0.5 atan2(sin2, cos2)
center = metric offset

Four projection axes · pred + GT
Lproj = mean[(1 − meanaxis GIoU)/2]
Corner distances: direct d0, reversed d180
Normalize by GT diagonal

Dpi = mean[min(d0, d180) / diagGT] · βcorner=1 · no extra geometry head

## Grouped objective

Car
Lcar

Ped / Cyc
Lvru

Ltotal = 0.5 Lcar + 0.5 Lvru

Masked Smooth-L1 divides by selected scalar coordinates + 1e−4; geometry averages positive boxes.

Empty targets: regression, geometry and IQA terms return differentiable zero; background focal supervision remains.